ANSWER KEY & MARKING SCHEME SET 1

Chapter 4: **Electricity: Magnetic and Heating Effects** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** cell or battery	A cell converts chemical energy to electrical energy.
2	**(a)** nichrome	Nichrome has high resistance and a high melting point.
3	**(a)** a resistor	Resistor symbol.
4	**(d)** a gap in the circuit	An open switch breaks the path.
5	**(d)** low melting point	It melts and breaks the circuit during overload.
6	**(a)** a broken connection or a fused filament	Open circuit.
7	**(d)** from the positive terminal through the circuit to the negative terminal of the cell	Conventional current is from the positive terminal.
8	**(c)** tungsten	Tungsten has a very high melting point.
9	**(a)** a thin nichrome wire	High resistance means more heating.
10	**(d)** copper	Metals conduct electricity.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	A closed path along which electric current flows (1); Includes a source, wires and a device (1).	1+1
12	It has a very high melting point (1); It can glow without melting (1).	1+1
13	High resistance gives more heat (1); High melting point and does not oxidise easily (1).	1+1
14	Conductor allows current to pass, e.g. copper (1); Insulator does not, e.g. rubber (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Cell, switch and bulb connected in a loop with wires (1); When the switch is closed the path is complete (1); Current flows through the filament and the bulb glows (1).	1+1+1
16	It is connected in series (1); Excess current heats and melts it (1); The circuit breaks and the appliances are safe (1).	1+1+1
17	Electric iron (1); Geyser, toaster, heater (1); Bulb filament (1).	1+1+1
	OR: Open switch makes a gap (1); The circuit is incomplete (1); No current flows (1).	1+1+1
18	Do not touch bare wires (1); Do not use wet hands on switches (1); Do not overload sockets (1).	1+1+1
	OR: To start or stop current (1); To save electricity (1); For safety (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Set-up: wire, cell, switch, compass (1); Needle deflects when current flows (1); Conclusion: current has a magnetic effect (1); Reverse current: deflection reverses (1); Needle returns when current is off (1).	1+1+1+1+1
	OR: Crane: current ON lifts scrap (1); Current OFF drops it (1); Magnet separates iron from rubbish (1); Soft iron is easily magnetised and demagnetised (1); Steel would stay magnetised (1).	1+1+1+1+1
20	Wind insulated wire around an iron nail (1); Connect to a cell (1); Test by lifting paper clips (1); More turns means more clips (1); More current means stronger (1).	1+1+1+1+1
	OR: Thin wires have more resistance (1); Nichrome has more resistance than copper (1); More resistance gives more heat (1); Thick copper wires carry current safely (1); So house wiring uses copper (1).	1+1+1+1+1

Mistakes to avoid:

- Saying the current direction does not matter for the compass needle; reversing it reverses the deflection.
- Using steel or plastic as the core of an electromagnet; soft iron is used for a temporary magnet.
- Choosing a fuse of very large rating or writing that a fuse wire has a *high* melting point.
- Forgetting that a circuit must be closed (switch ON) for current, and for the compass to react.
- Not insulating the coil wire, or forgetting to say that the magnetism ends when the current stops.

- Confusing the heating effect (bulb, heater, fuse) with the magnetic effect (electromagnet, compass).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 2

Chapter 4: **Electricity: Magnetic and Heating Effects** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** magnetic effect of current	An electromagnet pulls the hammer.
2	**(d)** magnetic effect of current	Oersted's observation.
3	**(d)** plastic	Plastic does not allow current.
4	**(c)** live and neutral wires touch	Very large current flows.
5	**(b)** increase the number of turns of the coil	More turns, stronger field.
6	**(c)** deflects a magnetic compass needle placed near it	Current produces a magnetic field.
7	**(c)** too many appliances connected at once	The total current becomes excessive.
8	**(c)** series with the appliance	To break the circuit.
9	**(a)** in the opposite direction	The direction of the field reverses.
10	**(d)** automatically when current exceeds a limit	It trips.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	It melts when current is too high (1); It protects appliances and prevents fire (1).	1+1
12	Live and neutral wires touch directly (1); A very large current flows (1).	1+1
13	A compass needle near a current-carrying wire gets deflected (1); Hence current produces a magnetic effect (1).	1+1
14	Electric iron (1); Geyser or heater (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Place a compass below a straight wire connected to a cell and a switch (1); Close the switch (1); The needle deflects; reversing the cell reverses the deflection (1).	1+1+1
16	Wires have some resistance (1); Electric energy changes to heat (1); Heat increases with the current (1).	1+1+1
17	Plastic and rubber are insulators (1); They prevent current passing to the user (1); Protect from shock (1).	1+1+1
	OR: Wind insulated wire on a soft iron nail and connect to a cell (1); Increase turns of the wire (1); Use more cells to increase the current (1).	1+1+1
18	Copper has low resistance (1); Nichrome has high resistance (1); Heat produced is more where resistance is high (1).	1+1+1
	OR: In both cases the needle deflects (1); Directions of deflection are opposite (1); Because the magnetic field direction differs above and below (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Electromagnet, armature, hammer, gong (1); Current magnetises the iron core (1); Armature attracted; hammer strikes gong (1); Contact breaks, current stops (1); Spring pulls it back and repeats (1).	1+1+1+1+1
	OR: Do not touch bare wires (1); Use proper plugs and insulators (1); Switch off before repair (1); Avoid water near sockets (1); Do not overload; use fuses (1).	1+1+1+1+1
20	Electric energy changes into heat (1); Heater, iron, geyser (1); Nichrome has high resistance (1); High melting point (1); Does not burn easily in air (1).	1+1+1+1+1
	OR: Reverse current changes deflection direction (1); Compass above or below gives opposite deflections (1); A larger current gives larger deflection (1); The field forms circles round the wire (1); Needle aligns with the field (1).	1+1+1+1+1

Mistakes to avoid:

- Saying the current direction does not matter for the compass needle; reversing it reverses the deflection.
- Using steel or plastic as the core of an electromagnet; soft iron is used for a temporary magnet.
- Choosing a fuse of very large rating or writing that a fuse wire has a *high* melting point.
- Forgetting that a circuit must be closed (switch ON) for current, and for the compass to react.

- Not insulating the coil wire, or forgetting to say that the magnetism ends when the current stops.
- Confusing the heating effect (bulb, heater, fuse) with the magnetic effect (electromagnet, compass).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 4: **Electricity: Magnetic and Heating Effects** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** electric iron	Iron heats a plate.
2	**(c)** soft iron core	Soft iron is strongly magnetised by current.
3	**(d)** plastic is an insulator	Prevents shocks and shorts.
4	**(d)** a combination of two or more cells	Cells joined.
5	**(c)** the current is switched off	Soft iron loses magnetism quickly.
6	**(a)** ampere	Symbol A.
7	**(d)** cranes that lift scrap iron	Magnets on and off.
8	**(b)** an electric heater	Current heats the high-resistance element.
9	**(d)** a long line and a short thick line	The long line is positive.
10	**(d)** switching off appliances before repair	Safety first.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Connecting too many appliances to one socket (1); Total current becomes dangerously high (1).	1+1
12	A coil carrying current wound around a soft iron core (1); It behaves like a magnet only while current flows (1).	1+1
13	Electric bell (1); Electromagnetic crane (1).	1+1
14	Water conducts electricity (1); Wet hands may cause an electric shock (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Permanent magnet has constant magnetism (1); Electromagnet works only when current flows (1); Strength of an electromagnet can be changed (1).	1+1+1
16	Fuse melts once, MCB switches off and can be reset (1); Fuse must be replaced (1); MCB is more convenient and faster (1).	1+1+1
17	Household appliances are connected in parallel (1); Each gets the same voltage (1); One can be switched off independently (1).	1+1+1
	OR: Current flows through the coil and magnetises the iron core (1); The core pulls the armature; the hammer strikes the gong (1); The circuit breaks and the process repeats (1).	1+1+1
18	Electric bell (1); Lifting heavy iron objects in cranes (1); Separating iron from scrap or ore (1).	1+1+1
	OR: Fused filament (1); Loose connection (1); Cell is used up (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Fuse wire melts when current is high (1); MCB trips automatically (1); Short circuit (1); Overloading (1); Both break the circuit (1).	1+1+1+1+1
	OR: Cell: long and short lines (1); Switch open: gap (1); Switch closed: joined (1); Bulb: circle with a cross (1); Resistor: zigzag (1).	1+1+1+1+1
20	Connect a material in the gap in a circuit (1); Conductor: bulb glows, e.g. copper, iron, aluminium (1); Insulator: bulb does not glow, e.g. rubber, plastic, wood (1); Metals conduct due to free electrons (1); Use in wires and handles (1).	1+1+1+1+1
	OR: Large current heats the wire (1); May melt insulation and cause fire (1); Use proper fuse or MCB (1); Do not connect too many appliances (1); Use good quality wires (1).	1+1+1+1+1

Mistakes to avoid:

- Saying the current direction does not matter for the compass needle; reversing it reverses the deflection.
- Using steel or plastic as the core of an electromagnet; soft iron is used for a temporary magnet.
- Choosing a fuse of very large rating or writing that a fuse wire has a *high* melting point.
- Forgetting that a circuit must be closed (switch ON) for current, and for the compass to react.

- Not insulating the coil wire, or forgetting to say that the magnetism ends when the current stops.
- Confusing the heating effect (bulb, heater, fuse) with the magnetic effect (electromagnet, compass).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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